# Risk Management

- A **risk** is any ***event or circumstance*** that could ***negatively impact*** the project, the software, or the organization.
- Different categories of risks

**a. Project risks**

- Threaten the ***project plan***
- If project risks become real, the schedule may slip, and costs may rise.
- Risks can affect the budget, timeline, staff, resources, stakeholders, and project requirements.

**b. Technical risks (Product risks)**

- Threaten the ***quality and timeliness*** of the software to be produced
- Technical risks make a project harder to complete because of design, implementation, or maintenance challenges.
- These risks can come from unclear requirements, outdated technology, or using new, untested methods.
- If a technical risk happens, it may slow down or stop the project.

### c. Business risks

- Business risks can threaten a ***software project and its success***.
- The top five business risks include:
  - ✓ **Market Risk** : Building a great product that no one wants.
  - ✓ **Strategic Risk:** The product no longer aligns with the company's goals.
  - ✓ **Sales Risk**: The sales team doesn't know how to sell it.
  - ✓ **Management Risk**: Losing senior management support.
  - ✓ **Budget Risk**: Losing funding or key team members.

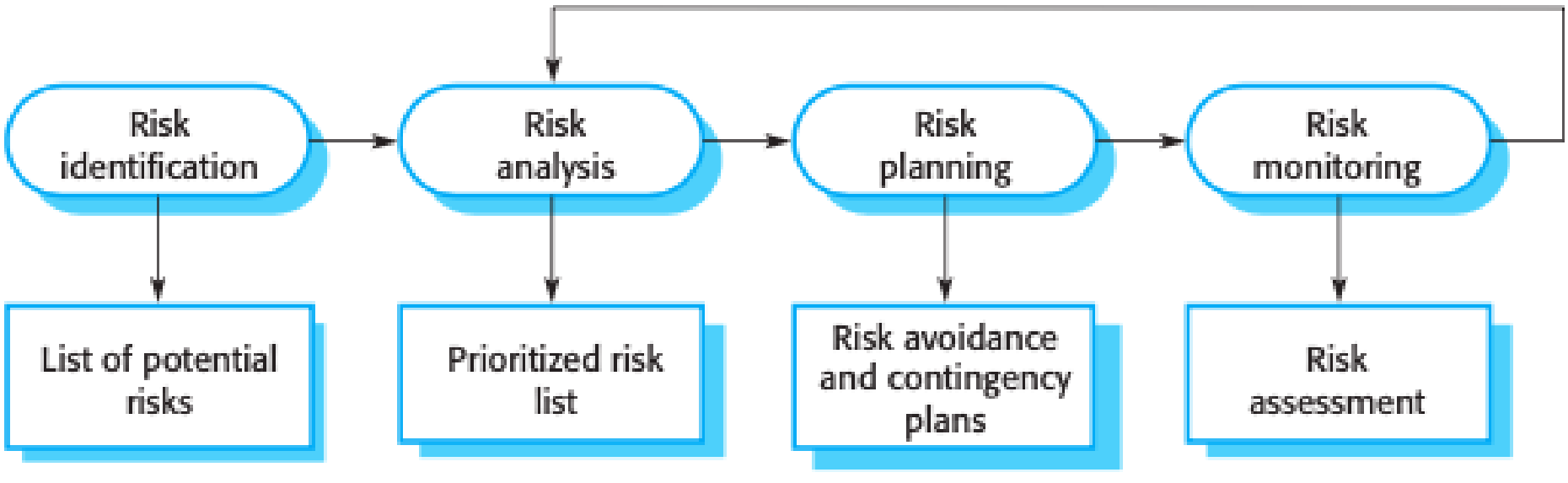
- **Known risks** are those identified through careful review of the project plan, business and technical environment, and reliable sources.
  - Examples include unrealistic deadlines, unclear requirements, and poor development conditions.
- **Predictable risks** come from past project experiences, like staff turnover, poor communication with customers, and team distractions from maintenance tasks.
- **Unpredictable risks** are like wild cards—they can happen anytime, but they're hard to foresee.

## Examples of common project, product, and business risks

| Risk | Affects | Description |
|---|---|---|
| Staff turnover | Project | Experienced staff will leave the project before it is finished. |
| Management change | Project | There will be a change of company management with different priorities. |
| Hardware unavailability | Project | Hardware that is essential for the project will not be delivered on schedule. |
| Requirements change | Project and product | There will be a larger number of changes to the requirements than anticipated. |
| Specification delays | Project and product | Specifications of essential interfaces are not available on schedule. |
| Size underestimate | Project and product | The size of the system has been underestimated. |
| Software tool underperformance | Product | Software tools that support the project do not perform as anticipated. |
| Technology change | Business | The underlying technology on which the system is built is superseded by new technology. |
| Product competition | Business | A competitive product is marketed before the system is completed. |

# Risk Management Process

- Risk management is an ongoing process throughout a project.
- After creating a risk plan, you must monitor for new risks, update priorities, and adjust plans as needed.



## 1. Risk Identification

- Risk identification is the first step in risk management.
- It involves ***finding potential risks*** that could affect the software development process, the software itself, or the organization.
- This can be done through **team brainstorming** or by **project managers** using their past experiences to identify common problems.
- Six types of risk may be included in a risk checklist:
  - **Estimation Risks**: Mistakes in estimating time and resources needed.
  - **Organizational Risks:** Issues within the company affecting development.
  - **People Risks:** Problems related to the development team.
  - **Requirements Risks:** Changes in customer needs and managing them.
  - **Technology Risks:** Challenges with software or hardware used.
  - **Tools Risks:** Problems with the software tools used for development.
- **Pruning the List**:
  - Necessary to focus on the most critical risks and avoid overwhelming the risk management process.

| Risk type | Possible risks |
|---|---|
| Estimation | 1. The time required to develop the software is underestimated.<br>2. The rate of defect repair is underestimated.<br>3. The size of the software is underestimated. |
| Organizational | 4. The organization is restructured so that different management are responsible for the project.<br>5. Organizational financial problems force reductions in the project budget. |
| People | 6. It is impossible to recruit staff with the skills required.<br>7. Key staff are ill and unavailable at critical times.<br>8. Required training for staff is not available. |
| Requirements | 9. Changes to requirements that require major design rework are proposed.<br>10. Customers fail to understand the impact of requirements changes. |
| Technology | 11. The database used in the system cannot process as many transactions per second as expected.<br>12. Faults in reusable software components have to be repaired before these components are reused. |
| Tools | 13. The code generated by software code generation tools is inefficient.<br>14. Software tools cannot work together in an integrated way. |

### a. Risk Components and Drivers

- The U.S. Air Force has published guidelines for identifying and reducing software risks.
- Their approach focuses on four key risk areas:
  - **Performance**: Ensuring the software meets functional and quality expectations.
  - **Cost**: Keeping the project within budget.
  - **Support**: Maintaining and updating the software efficiently.
  - **Schedule**: Completing the project on time.
- The project manager identifies factors (risk drivers) that could impact these areas and works to minimize risks.

## 2. Risk Analysis

- In the risk analysis process, you ***evaluate each identified risk*** based on two factors:

**(i) Probability Levels:**(Likelihood of occurrence)

- **Insignificant** (very unlikely)
- **Low** (unlikely)
- **Moderate** (possible)
- **High** (likely)
- **Very High** (almost certain)

**(ii) Impact Levels:**(Effect on the project)

- **Catastrophic** (project failure)
- **Serious** (major delays or cost overruns) or **Critical**
- **Tolerable** (manageable delays within contingency) or **Marginal**
- **Insignificant** (minimal or no effect) or **Negligible**

- By assigning each risk to these categories, you can prioritize which risks need the most attention and mitigation.

- Regularly update the risk table as new information emerges and management plans take effect.
- After ranking risks, focus on the most significant ones based on their probability and impact.
- Always consider catastrophic risks and serious risks with a moderate or higher chance of occurring.

| Risk | Probability | Effects |
|---|---|---|
| Organizational financial problems force reductions in the project budget (5). | Low | Catastrophic |
| It is impossible to recruit staff with the skills required (6). | High | Catastrophic |
| Key staff are ill at critical times in the project (7). | Moderate | Serious |
| Faults in reusable software components have to be repaired before these components are reused (12). | Moderate | Serious |
| Changes to requirements that require major design rework are proposed (9). | Moderate | Serious |
| The organization is restructured so that different managements are responsible for the project (4). | High | Serious |
| The database used in the system cannot process as many transactions per second as expected (11). | Moderate | Serious |
| The time required to develop the software is underestimated (1). | High | Serious |
| Software tools cannot be integrated (14). | High | Tolerable |
| Customers fail to understand the impact of requirements changes (10). | Moderate | Tolerable |
| Required training for staff is not available (8). | Moderate | Tolerable |
| The rate of defect repair is underestimated (2). | Moderate | Tolerable |
| The size of the software is underestimated (3). | High | Tolerable |
| Code generated by code generation tools is inefficient (13). | Moderate | Insignificant |

## Assessing Risk Impact

- Three key factors determine the consequences of a risk:
  - **Nature**: What kind of problem will arise if the risk happens?
  - **Scope**: How serious is the risk, and how widely does it affect the project or stakeholders?
  - **Timing**: When will the impact be felt, and for how long? Early detection is usually better, but sometimes delaying the issue is preferable.
- To assess a risk's overall impact, follow these steps:
  - ***Determine the average likelihood of occurrence.***
  - ***Use a risk impact chart to evaluate the effect of each component.***
  - ***Complete a risk table and analyze the results.***
- The overall risk exposure RE is determined using the following relationship

$$\boxed{RE = P \times C}$$

**P** : Probability of occurrence for a risk
**C** : Cost to the project should the risk occur.

Assume that the software team defines a project risk in the following manner:

- **Risk Identification:**
  - Only 70% of planned reusable components will work, meaning the remaining 30% must be custom-built.
- **Risk Probability:**
  - There is an 80% chance this will happen.
- **Risk Impact:**
  - 18 components (each 100 lines of code) must be developed from scratch.
  - Since each line costs $14 to develop,

$$\boxed{\textbf{Total cost impact} = 18 \times 100 \times 14 = 25{,}200}$$

- **Risk Exposure:**
  - Calculated as probability × impact:

$$\boxed{\begin{aligned} RE &= 0.80 \times 25{,}200 \\ &= 20{,}200 \end{aligned}}$$

- So, the expected cost due to this risk is **$20,200**.

## Hardware Delivery Delay

- **Risk Identification**:
  - The arrival of essential hardware may be delayed, affecting testing and integration.
- **Risk Probability:**
  - 50% chance.
- **Risk Impact:**
  - A two-week delay requires keeping three engineers at $3,000 each.

**Total cost impact = 3 × 3,000 = $9,000**

- **Risk Exposure:**

**RE = 0.50 × 9,000**
**= $4,500**

**Expected cost due to this risk: $4,500**

## 3. Risk Planning

- Risk planning helps ***manage potential problems*** in a project by developing strategies to reduce their impact.
- To do this, you ask "**what-if**" questions to anticipate risks and prepare solutions.
- Examples of risks include:
  - ***Several team members getting sick at once.***
  - ***Budget cuts due to economic downturns.***
  - ***Losing an expert on critical software.***
  - ***A supplier going out of business.***
  - ***Delays in receiving updated requirements from the customer.***
- There are three main ways to manage risks:

**a. Avoidance Strategies**:

- Reduce the chance of the risk happening (e.g., using reliable suppliers to avoid defective components).

**b. Minimization Strategies**

- Reduce the impact if the risk does happen (e.g., cross-training employees to handle staff illness).

**c. Contingency Plans**

- Have a backup plan in case the worst happens (e.g., securing alternative funding for financial risks).

## Strategies to help manage risk

| Risk | Strategy |
|---|---|
| Organizational financial problems | Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business and presenting reasons why cuts to the project budget would not be cost-effective. |
| Recruitment problems | Alert customer to potential difficulties and the possibility of delays; investigate buying-in components. |
| Staff illness | Reorganize team so that there is more overlap of work and people therefore understand each other's jobs. |
| Defective components | Replace potentially defective components with bought-in components of known reliability. |
| Requirements changes | Derive traceability information to assess requirements change impact; maximize information hiding in the design. |
| Organizational restructuring | Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business. |
| Database performance | Investigate the possibility of buying a higher-performance database. |
| Underestimated development time | Investigate buying-in components; investigate use of automated code generation. |

## 4. Risk Monitoring

- Risk monitoring means regularly checking if risks to your project, product, or business have changed.
  - **Check Likelihood & Impact**: See if each risk is more or less likely and if its effect has grown or shrunk.
  - **Look at Key Factors**: For example, more change requests might mean higher risk.
  - **Review Risks at Every Stage**: Regularly assess risks during project progress.
  - **Discuss in Management Reviews**: Go over major risks, update their status, and adjust plans if needed.

| Risk type | Potential indicators |
|---|---|
| Estimation | Failure to meet agreed schedule; failure to clear reported defects. |
| Organizational | Organizational gossip; lack of action by senior management. |
| People | Poor staff morale; poor relationships among team members; high staff turnover. |
| Requirements | Many requirements change requests; customer complaints. |
| Technology | Late delivery of hardware or support software; many reported technology problems. |
| Tools | Reluctance by team members to use tools; complaints about software tools; requests for faster computers/more memory, and so on. |

# RISK MITIGATION, MONITORING, AND MANAGEMENT PLAN

- Risk analysis in software projects helps teams develop strategies to ***handle risks*** effectively.
- There are three key aspects of risk management:

**1. Risk Mitigation**

- Risk Mitigation is a technique for ***avoiding risks*** (Risk Avoidance).
- The following are steps to take to reduce the risks:
  - ***Identifying the risk.***
  - ***Getting rid of the causes that lead to the production of risk.***
  - ***Controlling the relevant documents regularly.***
  - ***Conducting timely reviews to accelerate the process.***

If high staff turnover is a risk, you can:

- Identify and fix issues that cause employees to leave.
- Ensure knowledge is shared across the team.
- Assign backup staff for critical roles.

## 2. **Risk Monitoring**:

- Risk monitoring is an activity used to ***track a project's progress***.
- The following are the critical goals of the task.
  - ***Check if expected risks actually occurred.***
  - ***Verify if risk prevention steps were properly followed.***
  - ***Collect information to improve future risk assessments.***
  - ***Identify which risks caused specific problems in the project.***

For staff turnover, monitor:

- Team morale and job satisfaction.
- Compensation issues.
- Job opportunities elsewhere.

## 3. Risk Management & Contingency Planning

- Risk management helps prepare for potential problems in a software project.
- If risks are well managed early on, they are easier to control.
- However, if things go wrong, the project manager must handle the situation.
- A key tool in this process is the **risk register**, which helps identify and prioritize potential threats.
- The **80-20 rule** helps focus on the most critical risks—about 20% of risks cause 80% of project failures.

If key team members leave:

- Use backups and shared knowledge to continue work.
- Adjust schedules and refocus resources.
- Have departing employees transfer their knowledge before leaving.

- A risk management strategy can be part of the software project plan or documented separately in a Risk Mitigation, Monitoring, and Management (RMMM) plan.
- The RMMM plan ***records all risk-related activities*** and helps the project manager manage risks effectively.
- Some software teams do not create a formal Risk Mitigation, Monitoring, and Management (RMMM) document.
- Instead, they document each risk separately using a **Risk Information Sheet (RIS).**
- The RIS is usually stored in a database, making it easy to enter data, set priorities, search for risks, and analyze them.

| Risk information sheet | | | |
|---|---|---|---|
| Risk ID: P02-4-32 | Date: 5/9/09 | Prob: 80% | Impact: high |

**Description:**
Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.

**Refinement/context:**
Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards.
Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components.
Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.

**Mitigation/monitoring:**
1. Contact third party to determine conformance with design standards.
2. Press for interface standards completion; consider component structure when deciding on interface protocol.
3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.

**Management/contingency plan/trigger:**
*RE* computed to be $20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly.
Trigger: Mitigation steps unproductive as of 7/1/09.

**Current status:**
5/12/09: Mitigation steps initiated.

| Originator: D. Gagne | Assigned: B. Laster |
|---|---|

| **Risk ID:** RS1 | **Date:** 11/20/02 | **Prob**: 40% | **Impact:** Catastrophic | **Risk Value:** 1.6 |
|---|---|---|---|---|

**Description:**

Correctly importing Medical Database into the system is critical to getting symptoms, diagnosis and treatment. The database will need to be scrubbed and imported into tables.

**Refinement/context**

Subcondition 1: Database may not be in SQL standard format.

Subcondition 2: Database language not understood correctly by software engineer.

Subcondition 3: Database may not have direct symptom/diagnosis/treatment linkage necessary to support requirement R2.

**Mitigation/Monitoring**

1) Contact Database owner to determine format of database

2) If format is not SQL based, determine if conversion is possible to a relational database

3) Upon receipt of database from vendor, run preliminary data checks and reports to ascertain integrity of data

4) Have all members of project attend one day seminar on canine medical terms

5) Contact Database owner to determine if there is symptom/diagnosis/treatment linkage

**Management/Contingency Plan/Trigger**

If the understanding of the database is proceeding slowly (longer than a week), hire expert Veterinarian to help understand database. The risk exposure cost is approximately $2000 (RE=40%*$5000). $5000 is the cost of obtaining the services of veterinarian for 5 days. Allocate this amount into project contingency cost.

If transition of data over to new system database is bogged down, have 2 developers attend needed training to import non-standard formatted database into system database.